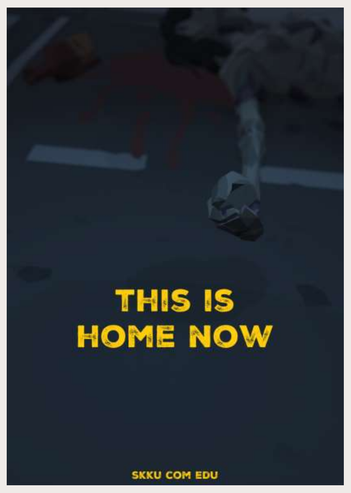
SUNG KYUN KWAN UNIVERSITY(SKKU)

This is Home Now

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Introduction





Title	This is home now
Environment	Unity 3D
Platform	PC
장르	Survival, Zombie, Open World
Reference	This is War of Mine, Project Zomboid, No Oxygen Included

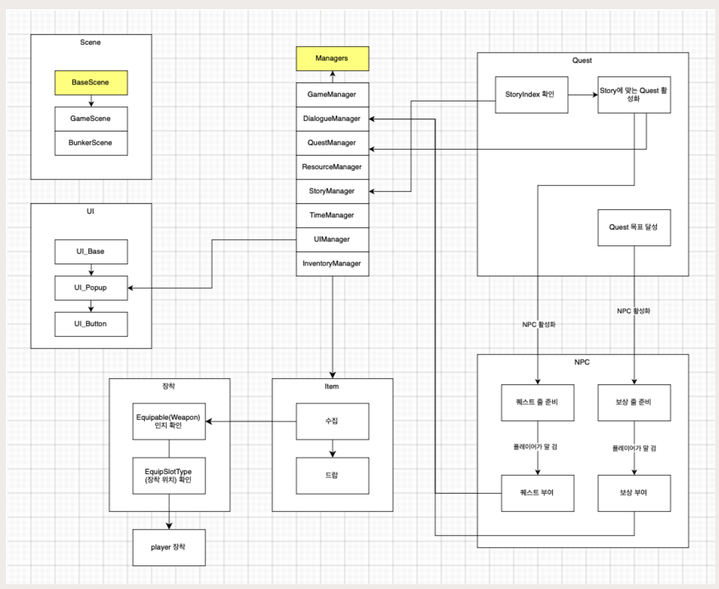
"Survive, Manage, and Optimize" Set in a post-apocalyptic zombie world, this simulation integrates two distinct gameplay loops: Strategic base management and high-stakes exploration. Survival relies on leveraging unique NPC traits, managing limited resources, and choosing avoidance over combat.
The game is designed to induce constant tension, compelling players to optimize their choices between resource consumption and risk-taking in a harsh environment.

Game Interface





Architecture



System



- Stealth-Oriented AI & Combat Design

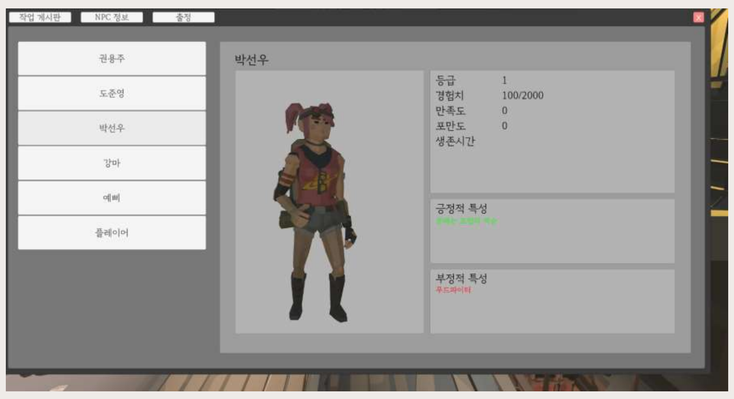
Designed to discourage direct combat. The AI utilizes Acoustic Sensitivity, where noise from firearms or skirmishes triggers a cascading swarm effect, drawing all nearby enemies to the sound source.
Combat balance follows a strict rule: "One is manageable, two is lethal, three is impossible." This forces players to prioritize stealth, evasion, and silence over brute force.
Zombies employ environment-aware AI that dynamically navigates terrain colliders to intercept players, requiring strategic movement to survive.

- AI Architecture & Optimization

Utilized Object Pooling to manage the instantiation and destruction of zombie entities dynamically. This minimizes memory allocation/garbage collection overhead during high-density swarms.
AI agents utilize dynamic collider detection to navigate complex terrain features realistically, ensuring fluid pursuit mechanics without getting stuck on geometry.

- NPC

Survival relies on "Needs-Driven" exploration—scavenging not for wealth, but for another day of life. The core tension, however, lies in human resources. NPCs are essential for optimizing bunker production, but they come with a twist: desperation masks defects. To secure a spot in the safe house, NPCs hide their negative traits, showing only their strengths. This forces players to make critical, intuitive decisions: is this survivor a vital asset, or a ticking time bomb?



- Feature & UX Focus

Interactive Item Management

A fully interactive inventory system allowing seamless item manipulation—equip, consume, or discard—via simple gestures.
Changes in the inventory are instantly reflected on the character model, bridging the gap between 2D UI and 3D gameplay.
Utilized Scriptable Objects to create a flexible and expandable item database, ensuring easy addition of new content without code refactoring.

- Events

The game features a dynamic event system where low NPC morale or specific dates trigger critical threats like mutiny or external invasions. This forces players into a strategic corner: should you drain your limited resources to guarantee safety, or gamble on luck to accelerate expansion? This mechanic transforms simple resource management into a tense psychological battle, challenging players to balance aggressive efficiency against the risk of catastrophic failure





- Real-Time Mass Rendering System Challenge

Rendering a dense open-world city with 10,000+ dynamic objects and zombie swarms caused severe CPU bottlenecks and unplayable frame rates using the standard rendering pipeline. For solution, architected a custom optimization pipeline utilizing GPU Instancing to merge thousands of draw calls for identical meshes into single batches. Implemented Multithreading to offload heavy AI and physics calculations from the main thread

Conclusion

This game maximizes tension by juxtaposing the constant threat of zombie hostilities against a desperate objective of escape. Strategic depth is amplified through dynamic NPC interactions, where securing survival tools and navigating crises require calculated decision-making. As a story-driven adventure, it deepens player immersion by integrating item-based environmental puzzles with exploration. The game’s architecture is designed for scalability, promising richer content through future genre hybridization that demands creative player agency. Furthermore, the development roadmap includes the potential integration of logic mechanics to foster critical thinking, positioning the game as a sustainable platform that balances visceral tension with intellectual engagement.